Exp No: 7 Random Sampling and Sampling Distribution

AIM:

To explore random sampling from a population and understand the concept of sampling distribution using Python.

PROGRAM:

```
import numpy as np

import matplotlib.pyplot as plt

population_mean = 50

population_std = 10

population_size = 100000

population = np.random.normal(population_mean, population_std,
population_size)

sample_sizes = [30, 50, 100]

num_samples = 1000

sample_means = {}

for size in sample_sizes:

  sample_means[size] = []

  for _ in range(num_samples):

    sample = np.random.choice(population, size=size, replace=False)

    sample_means[size].append(np.mean(sample))

plt.figure(figsize=(12, 8))

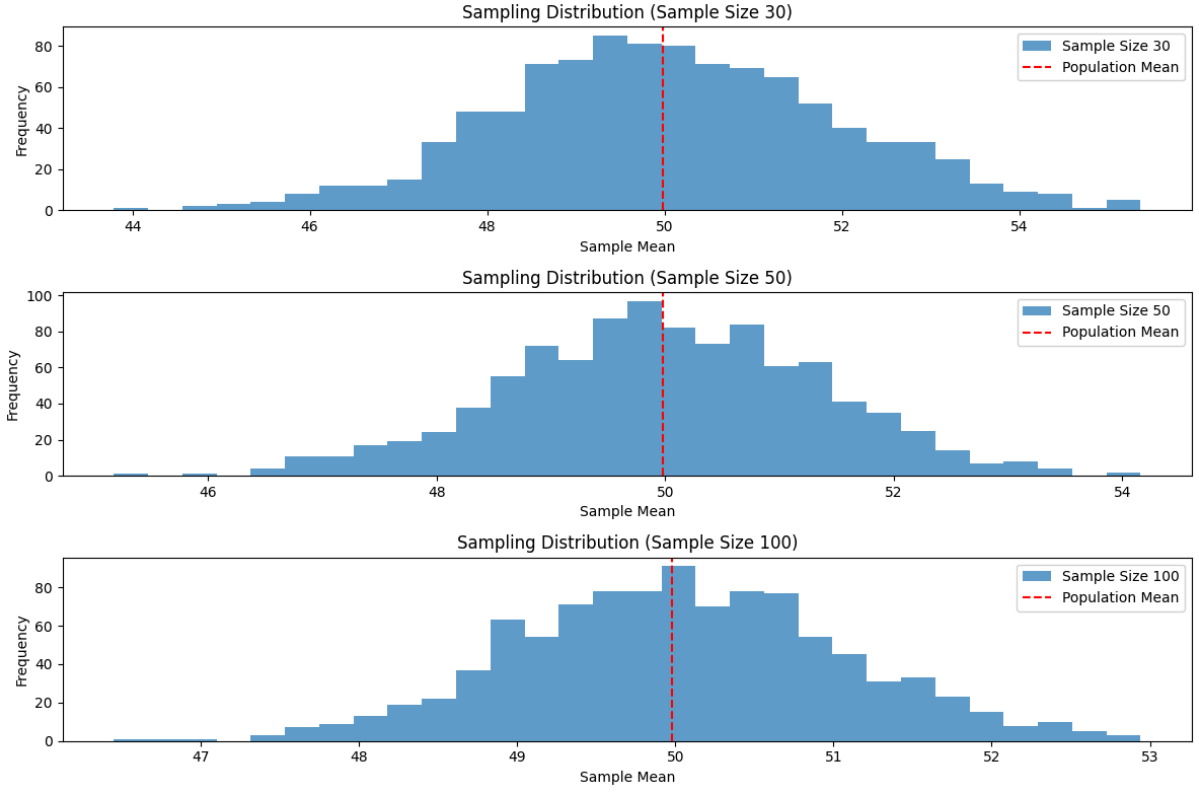
for i, size in enumerate(sample_sizes):

  plt.subplot(len(sample_sizes), 1, i+1)

  plt.hist(sample_means[size], bins=30, alpha=0.7, label=f'Sample Size {size}')
```

```
  plt.axvline(np.mean(population), color='red', linestyle='dashed',linewidth=1.5,
label='Population Mean')
  plt.title(f'Sampling Distribution (Sample Size {size})')
  plt.xlabel('Sample Mean')
  plt.ylabel('Frequency')
  plt.legend()
plt.tight_layout()
plt.show()
```

OUTPUT:



RESULT:

Thus the concept of random sampling distribution using Python was explored.